BinaryDemoire: Moiré-Aware Binarization for Image Demoiréing

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1. More Details about MABG Module

To minimize the computational overhead, our MABG module is not applied to all the convolution layers. Specifically, as a network with a three-layer encoder-decoder architecture, we apply the MABG module only in key operations such as upsampling, downsampling, and convolution layers responsible for channel adjustment between the encoder and decoder stages; these operations have the most significant impact on the overall architecture. For those convolutions in dense blocks and channel attention blocks, which typically constitute the majority of the network, applying such operations would introduce a large computational overhead with limited improvements in performance.

Based on this strategy, we apply the MABG module to approximately one-third of the convolution layers in the network. This approach effectively avoids a significant increase in computational cost while still achieving substantial improvements in performance. By strategically applying MABG only to crucial parts of the network, we ensure that the computational complexity is minimized without compromising the model's overall effectiveness.

We conducted extensive experiments by applying the MABG module to all layers of the network, as shown in the Tab. 1. The results clearly indicate that using MABG in all layers does not lead to significant improvements in performance. In fact, it introduces a higher computational cost due to the increase number of floating-point operations. This suggests that our selective application of MABG is well-optimized, achieving a good balance between performance enhancement and computational efficiency.

2. More Visualizations And Analyses

In this section, we present additional visualizations and analyses to further illustrate the performance of our MABG and SGRA modules. First, we provide further visualizations of the activation distributions, both with and without the application of interleaved mixing. Next, we showcase more feature visualizations to better observe the specific effects of the MABG module on the network's behavior.

Table 1. Ablation on MABG .

METHOD	PSNR	SSIM	LPIPS
w/o MABG	20.68	0.7820	0.2912
Part (ours)	20.97	0.7856	0.2845
All MABG	20.97	0.7864	0.2821

2.1. Activation Distribution After SGRA

Fig. 1 presents distributions of the first three groups of features from convolution modules located in the encoder and decoder before and after applying interleaved mixing. The results illustrate how channel distributions evolve due to the interleaved mixing operation in the SGRA module. The distributions show that the interleaved mixing effectively reduces the differences between the groups, facilitating more balanced information exchange. As seen in the figure, this operation significantly improves the feature integration and helps enhance the overall performance by ensuring better transmission of full-precision information between groups.

2.2. Visualization Of Intermediate Features

We present more feature visualizations to observe the effects of MABG. The images show a comparison of intermediate features at the same position, with and without applying the MABG module. The MABG module assigns different gating weights to each channel, enhancing the contribution of strong features while suppressing weaker ones. As shown in the images, after applying MABG, the features contain more high-frequency information, and the noise is significantly reduced. This demonstrates how MABG helps preserve important details and reduces the information loss caused by binarization, which is crucial for tasks such as moiré removal. The results highlight the effectiveness of MABG in improving the feature quality for binarized networks.

3. More Visual Comparisons.

In this section, we present additional visual results comparing our approach with other state-of-the-art binarized methods as well as full-precision models. These visualizations highlight the superiority of our method in effectively capturing fine details and handling challenging artifacts, particularly in complex image structures. By showcasing the detailed comparison across various image categories, we illustrate how our model excels in preserving high-frequency

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